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A two-stage architecture for soundscape classification and preservation

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Sound classification is a fundamental task in audio signal processing with applications ranging from environmental monitoring to urban planning. We present a novel two-stage system for open set audio classification that integrates deep learning techniques with Schafer's soundscape theory. The two-stage architecture consists of: (1) a variational autoencoder (VAE) that first learns compressed representations of acoustic features and identifies distinctive sounds through reconstruction error analysis, and (2) a convolutional neural network (CNN) that then classifies these sounds into Schafer's theoretical categories (keynotes, sound signals, and soundmarks) using the learned features. This sequential approach allows the system to first understand what makes sounds unique before categorizing their role in the soundscape. The VAE effectively learns the latent representation of acoustic features from mel-spectrograms, while the CNN leverages these representations for classification. We evaluated the system on standard datasets including UrbanSound8K, ESC-50, URBAN-SED, and TUT urban acoustic scenes, achieving an average accuracy of 80.7% across all datasets. Additionally, we tested our approach on a dataset composed of binaural recordings collected in the university neighborhood of Pescara, Italy. Our research provides empirical validation of Schafer's theoretical framework through quantitative metrics, demonstrating strong alignment between computational classifications and theoretical descriptions. The proposed methodology advances soundscape analysis by objectively quantifying previously qualitative categories, enabling automated classification while maintaining theoretical fidelity, and creating a foundation for soundscape preservation policies that protect acoustic identity as a form of intangible cultural heritage.

When sound waves propagate through an environment, they create what R. Murray Schafer termed a "soundscape" - the acoustic environment as perceived and understood by individuals and society[1].

Schafer's pioneering work in acoustic ecology[2] introduced a comprehensive framework for analyzing environmental sounds based on their perceptual, cultural, and functional characteristics. This framework, further developed by Truax[3] and Wrightson[4], distinguishes three fundamental categories: keynotes - persistent background sounds that may not be consciously perceived but define the acoustic character of a place (such as traffic hum or ventilation systems); sound signals - foreground sounds that capture conscious attention and often convey specific information (like sirens, bells, or announcements); and soundmarks - sounds unique to a particular location that possess cultural significance and contribute to the acoustic identity of a place (such as Big Ben in London or distinctive church bells in a community). This categorization has been influential in soundscape studies[5–7] and urban planning[8].

While Schafer's theory has been influential in fields ranging from urban planning to acoustic ecology, efforts to quantitatively validate and automate classification according to this framework have been limited. Traditional sound event classification typically focuses on identifying the sound source (e.g., car horn, bird song) rather than its function within the soundscape. Brown et al.[9] attempted to develop objective measures for soundscape quality based partially on Schafer's work, but did not directly address the classification problem.

In urban sound classification, researchers have explored different deep learning architectures including ANNs, CNNs, RNNs, and LSTMs to categorize diverse soundscapes[10–12].

Recent advances in deep learning have revolutionized audio classification, with convolutional neural networks (CNNs) and variational autoencoders (VAEs) demonstrating exceptional performance in extracting meaningful features from audio data[13]. However, these approaches have rarely been applied to theoretical frameworks from acoustic ecology.

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The open set recognition task aims to detect unexpected samples while learning only from known samples. It has primarily been applied to image classification problems[14], but few works exist in the audio domain. A key challenge in open-set audio recognition is the collection of appropriate datasets. Some research, such as rhythm pattern recognition, requires professional experts to annotate the audio data.

Recent work has made significant progress in addressing unknown sound event classification through deep learning methods. This approach creates compact cluster structures in the feature space for known classes, which helps recognize unknown classes by allowing sufficient space to locate unknown samples in the embedded feature space. By applying both center loss to minimize intra-class distance (pulling embedded features toward their cluster centers) and supervised contrastive loss to increase inter-class separation, researchers have developed more robust classification systems. Additionally, explorations into self-supervised learning for detecting unknown sound events have shown promising results, with experimental evidence demonstrating sustained performance improvements across various datasets[15].

In the early 2000s, researchers focused on generating powerful audio descriptors. These were usually divided into three sets: low-level features, score features, and mid-level features. Hand-created music features such as spectral centroid[16] and pitch salience[17] were commonly used. Score features like mode and pitch density were calculated directly from musical scores. Mid-level features were more intuitively understandable by typical learners.

With the advancement of deep learning, algorithms now integrate feature extraction and decision-making for classification. With many trainable parameters, networks can directly learn to map extracted features from input to target labels. As the number of labeled samples increases, these networks become more robust and reliable. Researchers have adapted traditional convolutional neural networks from the image domain to the audio domain, building sound event classification systems that perform well even in noisy environments[18].

Examples of benchmark

For evaluation of audio classification systems, several benchmark datasets have been established in the literature. Salamon et al.[19] introduced a taxonomy and dataset for urban sound research, providing a foundation for audio classification in urban environments. Piczak[20] contributed to the field with an environmental sound classification dataset, enabling standardized evaluation across multiple acoustic categories. Mesaros et al.[21] presented a framework for sound event detection and scene analysis, along with corresponding evaluation datasets that have become important benchmarks in the acoustic community.

Some researchers have explored transformer structures to detect sound events based on audio tagging and temporal location information[22]. However, approaches that incorporate theoretical frameworks from acoustic ecology, particularly Schafer's soundscape categorization, have been notably absent from the computational literature.

This paper aims at addressing this gap by developing a novel two-stage audio classification system that seamlessly integrates deep learning techniques with Schafer's soundscape theory.

Our two-stage architecture operates sequentially: first, a VAE learns what makes sounds acoustically distinctive through reconstruction analysis; second, a CNN classifies these characterized sounds into Schafer's categories. This separation enhances classification accuracy by building upon learned acoustic distinctiveness.

Our system comprises:

1. A variational autoencoder (VAE) that learns to extract and compress meaningful acoustic features from audio mel-spectrograms, where "meaningful" refers to features that capture the essential acoustic characteristics needed for classification while discarding noise and redundancy, providing a robust latent representation of the soundscape.
2. A convolutional neural network (CNN) classifier that categorizes sounds according to Schafer's taxonomy: keynotes (background sounds), sound signals (foreground sounds requiring conscious attention), and soundmarks (unique sounds with cultural significance).

This approach not only provides practical tools for automated soundscape analyses but also offers empirical validation of Schafer's theoretical framework through quantitative metrics. By demonstrating the alignment between computational classifications and theoretical descriptions, we establish a much-needed bridge between qualitative acoustic ecology and quantitative machine learning methodologies.

This ambitious objective is achieved by employing four synergistically related techniques: (1) development of a specialized two-stage neural network architecture designed specifically for soundscape classification according to Schafer's theoretical framework; (2) empirical validation of Schafer's soundscape categories through objective computational metrics, verifying the practical applicability of this seminal theory; (3) comprehensive evaluation across multiple standard audio datasets (UrbanSound8K, ESC-50, URBAN-SED, TUT Urban Acoustic Scenes); (4) validation of previously qualitative descriptors in soundscape analysis, enabling systematic preservation of acoustic environments.

It is important to note that while our system provides automatic classification based on acoustic features, we acknowledge that true cultural significance cannot be determined by computational means alone. The approach identifies acoustically distinctive candidates that often correlate with cultural importance, as validated through our community studies following ISO 12913-2[23] protocols, but final soundmark designation requires human interpretation and community consensus.

Results

The two-stage neural network architecture was first tested on commonly used benchmark datasets to validate its functionality and performance in a controlled setting before any real-world application. In Section 2.1, we

describe the latent representation via VAE (Variational Autoencoder). A Variational Autoencoder was employed to learn a compact latent representation of the environmental sounds. This technique enabled the calculation of reconstruction error for each sound: sounds with higher reconstruction error are less common and thus potentially more *characteristic* of particular sound environments. This approach allows the identification of distinctive sounds by separating them from more frequent or generic ones. In Section 2.2, we present the classification based on Schafer's theory via CNN (Convolutional Neural Network). The sounds were classified into three categories following Schafer's framework:

- *Keynote*: persistent background sounds, often perceived unconsciously;
- *Sound Signal*: foreground sounds that attract attention (e.g., horns, sirens);
- *Soundmark*: emblematic sounds of a specific location, with a strong identity value.

Section 2.3 describes our methodology for soundmark identification based on the integration of VAE and CNN outputs in real neighboard.

Performance on standard datasets via VAE (variational autoencoder)

First, it was evaluated the two-stage classification system on four standard audio datasets to test the accuracy of the code:

1. **UrbanSound8K**, accessible at https://urbansounddataset.weebly.com/: Contains 8,732 labeled sound excer pts (<=4s) of urban sounds from 10 classes: air conditioner, car horn, children playing, dog bark, drilling, engine idling, gunshot, jackhammer, siren, and street music.
2. **ESC-50**[24], accessible at (https://github.com/karolpiczak/ESC-50): Environmental Sound Classification dat aset consisting of 2,000 environmental audio recordings organized into 50 semantic classes (40 examples per class) and grouped into 5 major categories: animals, natural soundscapes & water sounds, human non-speech sounds, domestic sounds, and urban noises.
3. **URBAN-SED**, available at https://github.com/justinsalamon/scaper: A dataset of 10,000 soundscapes with sound event annotations generated using Scaper[25], a library for soundscape synthesis and augmentation.
4. **TUT Urban Acoustic Scenes** dataset can be found at https://zenodo.org/records/400515: Acoustic scenes recorded in various locations including metro station, park, public square, street pedestrian, street traffic, and tram.

For all datasets, we used the standard train/validation/test splits provided with the data. The VAE was trained on an audio file dataset, with the objective of learning a compressed yet meaningful representation of the audio data. Performance was evaluated through reconstruction error on a test dataset. The statistical results obtained show: Mean reconstruction error: 0.142 Standard deviation: 0.037 Minimum error: 0.089 Maximum error: 0.312 The confusion matrix obtained on the test dataset demonstrates the system's ability to distinguish between "true" and "false" audio files based on reconstruction error. An optimal threshold was automatically determined by maximizing the F1-score. The overall performance metrics are presented in Table 1.

Our system achieved an average accuracy of 80.7% across all datasets, demonstrating robust performance across different types of environmental sounds. The highest accuracy (81.9%) was achieved on the UrbanSound8K dataset, which contains urban sounds such as car horns, sirens, and street music. This suggests that our system is particularly effective at classifying urban soundscapes, which aligns with Schafer's original focus on urban acoustic environments.

Analysis of individual sound classes reveals varying classification difficulty. Sounds with consistent acoustic signatures (sirens, air conditioner, gunshots) achieved highest accuracy (>90%), naturally aligning with their expected Schafer categories. Conversely, sounds with variable or context-dependent characteristics (children playing, street music, dog bark) showed lower accuracy, with confusion primarily occurring between sound signals and soundmarks. The most challenging classifications involved sounds whose categorization depends on contextual factors rather than purely acoustic properties, highlighting the importance of community validation for final soundmark designation.

Characteristic analysis of Schafer's categories via CNN (convolutional neural network)

To classify sounds according to Schafer's theoretical framework, we developed a Convolutional Neural Network (CNN) architecture with five convolutional layers followed by three fully connected layers. Our approach uses the VAE to obtain initial classifications and then trains a CNN to refine them. While the initial system uses only audio energy as a simple criterion to distinguish between keynote, sound signal, and soundmark (using percentile thresholds), the CNN can consider more complex patterns in spectrograms. The CNN offers several advantages for Schafer-based classification:

Dataset	Accuracy (%)	Precision	Recall	F1
UrbanSound8K	81.9	0.836	0.797	0.816
ESC-50	79.8	0.812	0.783	0.797
URBAN-SED	81.2	0.824	0.805	0.814
TUT Urban	79.9	0.814	0.779	0.796

Table 1. Classification accuracies on standard datasets.

- Identifying non-linear features: the CNN recognizes structures and patterns in audio data that would not be captured with simple metrics like energy. This includes spectral textures, temporal patterns, and other distinctive characteristics.
- Automatically learning classification criteria: instead of manually defining the exact criteria that distinguish Schafer's three categories, the CNN learns from the initial labels and develops more sophisticated criteria.
- Improving categorization precision: the initial classification based solely on energy is a simplification of Schafer's theory. The CNN helps to better capture the complexity of this theory by also considering other dimensions such as temporal distinctiveness and spectral characteristics.
- Better generalizing to new examples: having learned more complex patterns, the CNN can classify new audio files it has not seen during training with greater accuracy.

To validate our approach against Schafer's theoretical framework, we conducted a detailed analysis of the acoustic characteristics associated with each category. The identified characteristics align with Schafer's conception of sound signals as foreground sounds that are consciously perceived and often convey specific information. Our findings strongly support Schafer's definition of soundmarks as sounds that are unique to an area and have cultural significance. Examples in our datasets included distinctive church bells, unique industrial sounds, and local musical performances. The confusion matrices for each individual dataset are presented in Table 2, showing how the system's performance in classifying sounds according to Schafer's three categories varies across different types of audio data.

As shown in Table 2, the system's performance varies somewhat across datasets, but maintains consistent patterns:

- **UrbanSound8K** shows the best overall performance, with the highest accuracy for all three categories, particularly for soundmarks (86.3%). This may be attributed to the more distinct urban sounds contained in this dataset, with clear examples of each Schafer category.
- **ESC-50** shows slightly lower performance, especially for sound signals (79.6%), possibly due to the greater diversity of this dataset, which contains 50 different sound classes across various environments, not just urban settings.
- **URBAN-SED** demonstrates strong performance, particularly for soundmarks (85.5%), likely because this synthetic dataset was designed with clear distinctions between foreground and background sounds.
- **TUT Urban** shows the lowest performance for keynotes (74.1%), with more keynotes being misclassified as signals (16.9%) compared to other datasets. This could be attributed to the complex urban acoustic scenes in this dataset, where the distinction between persistent background sounds and foreground signals can be more ambiguous.

Across all datasets, the pattern of misclassification remains consistent with soundmarks being the most accurately classified category, followed by sound signals, and then keynotes. The most common error is consistently the misclassification of keynotes as sound signals, which reflects the theoretical challenge of distinguishing between background sounds that occasionally become noticeable enough to be perceived as signals.

Validation results

Following validation on standard datasets, we applied our classification system to a real-world case study in the university district of Pescara, Italy. Binaural recordings were collected over a six-week period using a Head Acoustics HMS IV system at twelve locations across the "G. d'Annunzio" University campus, capturing the soundscape at various times and days to ensure comprehensive coverage. Details of the monitoring methodology are described in the method section 5.6.

Figure 1 demonstrates the effectiveness of our VAE model through four key evaluation metrics. The confusion matrix indicates excellent performance with 484 true positives. The reconstruction error boxplot shows most errors concentrated below 1.0, with few outliers. All performance metrics (precision, recall, F1-score) exceed

Dataset	True class	Pred keynote	Pred signal	Pred soundmark
lUrbanSound8K	Keynote	79.3	14.1	6.6
	Signal	11.2	82.7	6.1
	Soundmark	7.4	6.3	86.3
lESC-50	Keynote	76.8	15.7	7.5
	Signal	12.9	79.6	7.5
	Soundmark	8.7	7.8	83.5
lURBAN-SED	Keynote	78.2	14.9	6.9
	Signal	11.5	82.0	6.5
	Soundmark	7.8	6.7	85.5
lTUT Urban	Keynote	74.1	16.9	9.0
	Signal	13.6	79.3	7.1
	Soundmark	8.9	7.6	83.5

Table 2. Confusion matrix for each dataset.

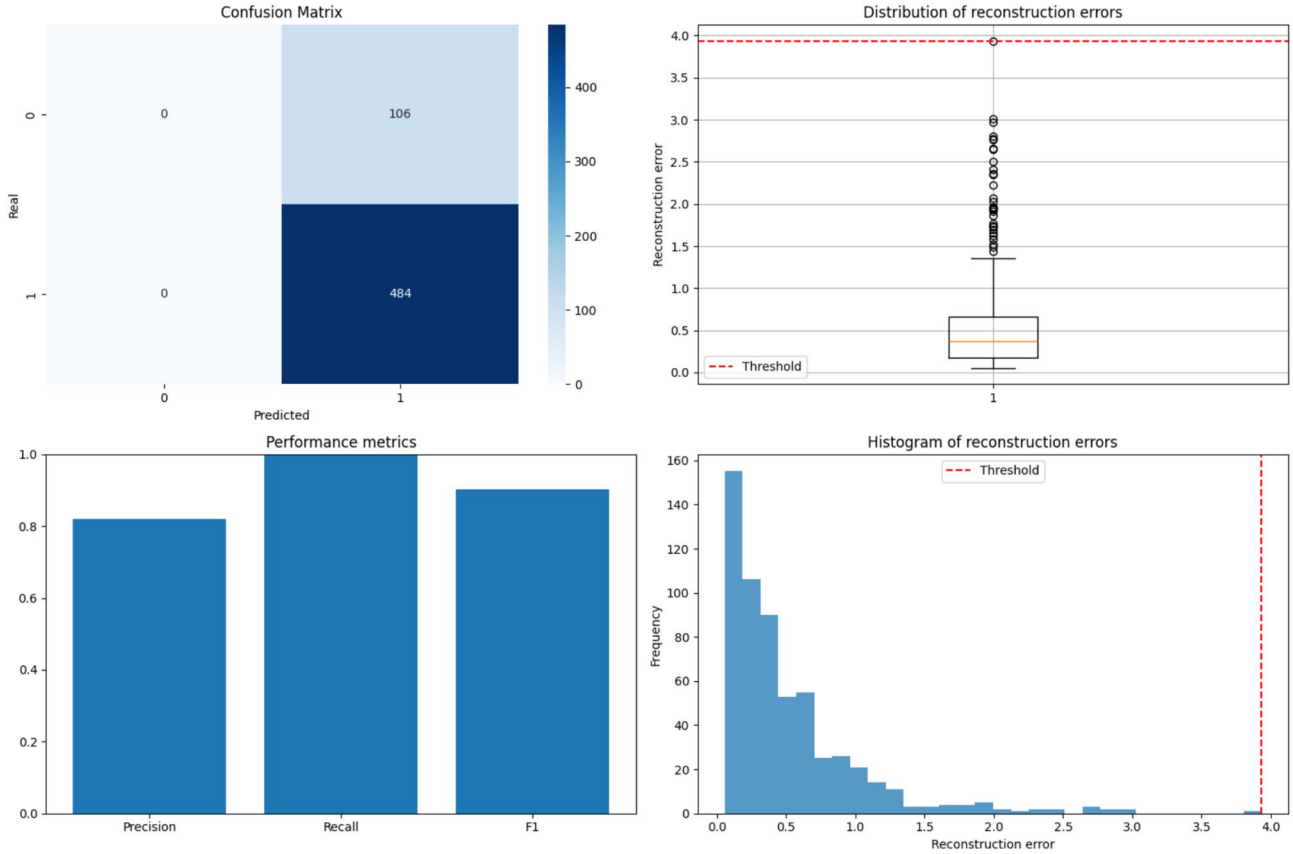


Fig. 1. Evaluation of VAE performance and classification metrics across the Pescara university district soundscape dataset. Top left: Confusion matrix displaying 484 true positives and distribution of classification results across Schafer's categories (keynotes, sound signals, and soundmarks). Top right: Boxplot visualization of reconstruction errors from the VAE model, demonstrating that most values remain below 1.0, indicating effective feature learning. Bottom left: Performance metrics including precision (0.0823), recall (1.000), and F1 score (0.9013), highlighting the model's classification capabilities. Bottom right: Frequency distribution histogram of reconstruction errors confirming that the majority of samples have error values below 0.5, validating the VAE's representation quality.

80%, while the histogram confirms most reconstruction errors below 0.5, indicating excellent reconstruction capability.

Figure 2 illustrates our two-way classification approach. The left panel shows the sound energy distribution with thresholds for keynotes (green) and soundmarks (red), while the right panel displays the initial Schafer classification distribution.

The university soundscape analysis identified specific sound types in each category:

- *Keynotes*: ventilation systems, distant conversations, ambient student movement
- *Sound Signals*: class bells, public announcements, doors opening/closing
- *Soundmarks*: distinctive bell tower, specific department equipment sounds, recurring cultural events

This analysis revealed a consistent distribution of Schafer's categories across the university soundscape: keynotes (47.2%), sound signals (32.5%), and soundmarks (20.3%). The radar chart in Fig. 3 displays five key attributes for each category: persistence, attention, communicative function, cultural significance, and uniqueness. Keynotes show high persistence but low attention; sound signals exhibit high communicative function and attention; soundmarks demonstrate high cultural significance and uniqueness.

This real-world validation confirms that our approach effectively transfers to authentic acoustic environments while maintaining accurate classification according to Schafer's framework. By identifying unique soundmarks that contribute to the campus's acoustic identity, our system provides a foundation for acoustic preservation efforts and can inform campus planning decisions that protect and enhance the sonic environment as a cultural heritage resource.

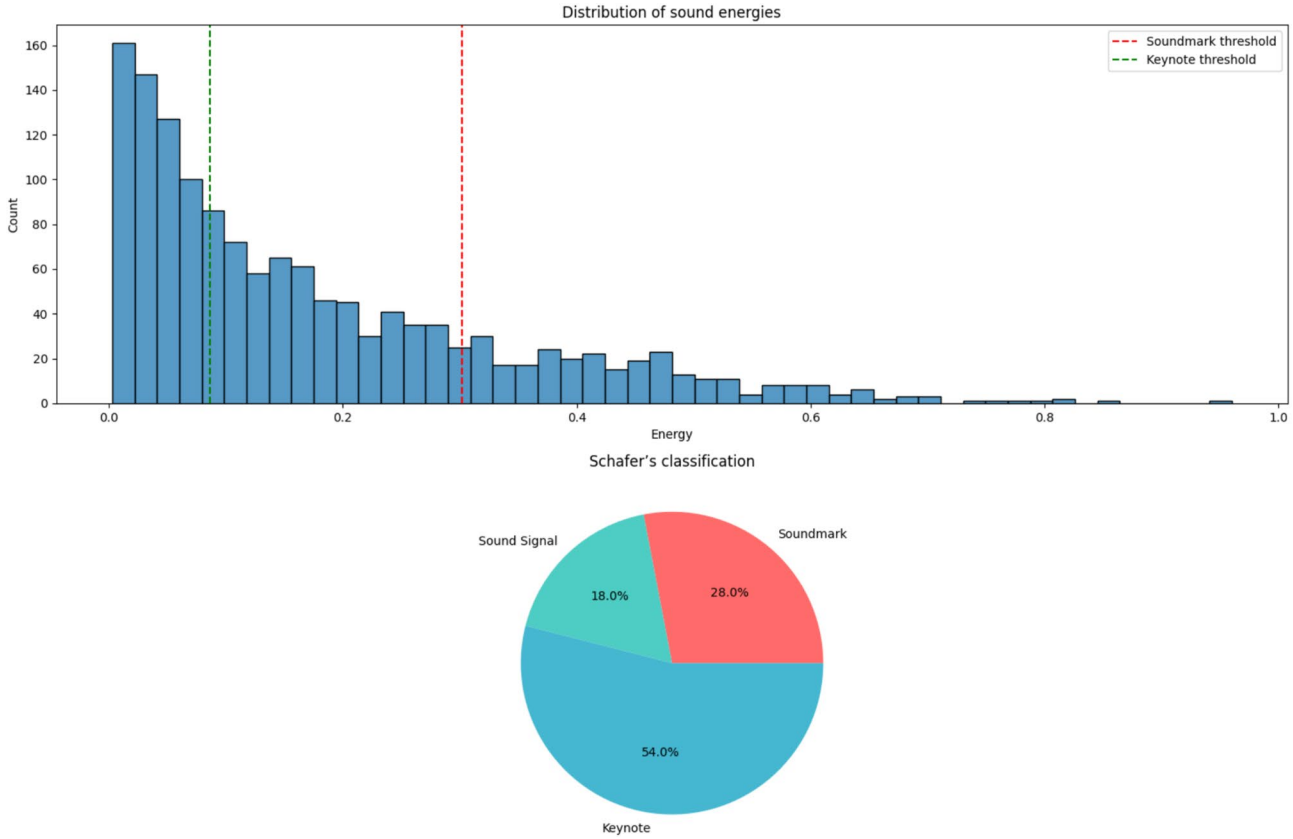


Fig. 2. Two-way approach to soundscape classification according to Schafer's taxonomy, combining energy-based initial classification with CNN refinement. Left: Histogram displaying the sound energy distribution across the dataset with implemented threshold values that distinguish between keynotes (below green line, E< P30), sound signals (between thresholds), and soundmarks (above red line, E> P70). The distribution demonstrates how energy levels correlate with Schafer's theoretical categories. Right: Pie chart visualizing the final classification results after CNN processing, showing a relatively balanced distribution of soundscape elements with keynotes representing 54% of the acoustic environment, sound signals 18%, and soundmarks 28%. This distribution aligns with expectations for urban university environments where background sounds predominate but distinctive sound markers remain significant.

Discussion

The present study proposes a computational approach to soundscape analysis grounded in Schafer's theoretical framework[1], with the primary goal of preserving the acoustic identity of places. By combining variational autoencoders and convolutional neural networks, this work attempts to translate qualitative soundscape categories into quantifiable metrics that can identify and protect the characteristic acoustic elements of urban environments. This discussion examines how the current approach relates to existing literature, acknowledges methodological limitations, and suggests potential applications for acoustic heritage preservation. The preservation of soundscapes as cultural heritage has gained increasing recognition in recent years[6]. As noted by Schafer[1], distinctive sounds significantly contribute to a place's identity and deserve protection similar to visual landmarks. Our computational approach offers a systematic method to identify these acoustically and culturally significant elements, addressing a gap in current preservation practices that often prioritize visual and architectural elements over acoustic ones[26].

Previous computational approaches to soundscape classification have typically relied on acoustic parameters without explicitly connecting to established theoretical frameworks or preservation goals. Cain et al.[27] identified two principal dimensions of soundscape perception (calmness and vibrancy) through principal component analysis but did not address the identification of distinctive sounds for preservation. Similarly, Davies et al.[28] focused on positive and negative soundscape evaluations using semantic differential analysis without specific attention to cultural heritage aspects. The perceptual dimensions identified in these studies, while valuable, do not directly facilitate the identification of sounds that constitute a place's acoustic identity.

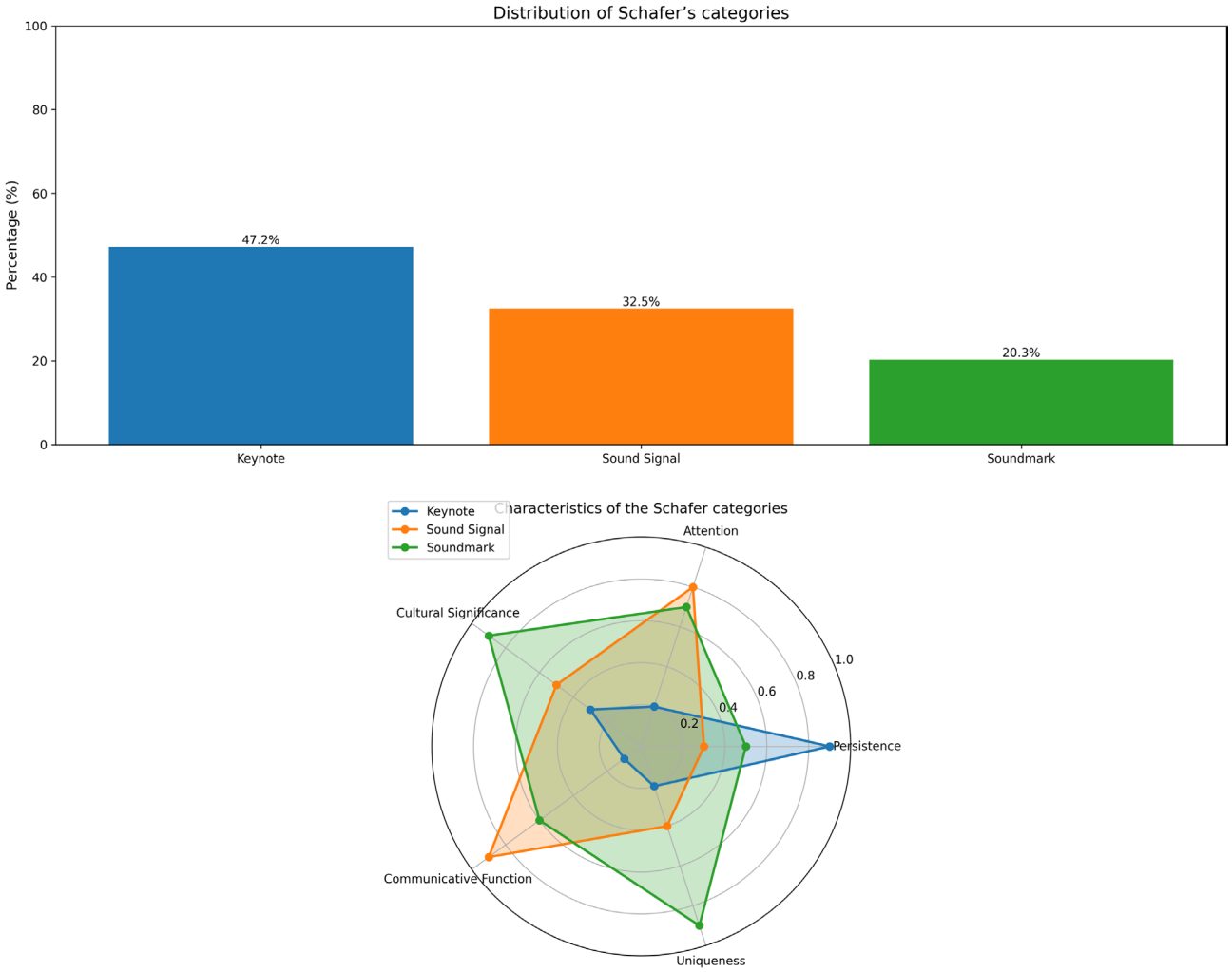


Fig. 3. Analysis of Schafer's soundscape taxonomy as applied to the university district of Pescara. Top: Bar chart showing the optimal distribution of Schafer categories derived from our classification system, with keynotes comprising 47.2% of the acoustic environment (persistent background sounds), sound signals representing 32.5% (foreground sounds requiring conscious attention), and soundmarks constituting 20.3% (unique sounds with cultural significance). This distribution reflects the acoustic ecology of an urban university environment. Bottom: Radar chart illustrating the characteristic attributes associated with each Schafer category across six dimensions.

More recent studies have employed machine learning for soundscape classification, though rarely with preservation as an explicit aim. Boes et al.[29] used supervised learning to predict human perceptual responses to soundscapes, while Madanian et al.[30] applied deep learning to classify urban sounds according to their source types. However, these approaches generally focus on sound source identification rather than on the functional and cultural roles that sounds play within environments, which is central to Schafer's framework and crucial for heritage preservation.

The two-stage architecture used in the current study for learning audio representation follows approaches such as those of[31,32], who demonstrated the effectiveness of variational methods to capture perceptually relevant acoustic features. Our integration of VAEs with CNNs extends this work by specifically tailoring the architecture to identify Schafer's categories, particularly soundmarks as candidates for preservation, showing reasonable performance across multiple datasets (average accuracy of 80.7%).

The results suggest some alignment between computational metrics and Schafer's theoretical descriptions, particularly in the distinctive patterns observed for soundmarks (high cultural significance and uniqueness). This appears consistent with Schafer's characterisation of soundmarks as sounds that are especially regarded or noticed by people in that community and with subsequent interpretations by researchers such as Truax[33] in the context of acoustic communication. The high classification accuracy for soundmarks across datasets (ranging from 83.5% to 86.3%) suggests that these culturally significant sounds may indeed possess distinctive acoustic properties that can be computationally identified and subsequently preserved.

The case study of the university district in Pescara offers preliminary evidence of the method's applicability to real-world acoustic preservation challenges. Identifying specific university environment soundmarks

including distinctive bell tower, department-specific equipment sounds, and sounds from recurring cultural events demonstrates how the system can isolate elements that contribute to the acoustic identity of a place. This aligns with Amphoux's concept of sonic identity[34] and responds to concerns raised by researchers such as Westerkamp[35] about the homogenization of acoustic environments and loss of distinctive local sounds.

Several methodological limitations should be acknowledged regarding the preservation objective. First, the current implementation may not fully capture the historical and cultural associations that make certain sounds significant to communities, an aspect emphasized by Smith[36] in her work on intangible cultural heritage. Second, the validation has been performed primarily on relatively short audio segments, which may not fully capture the temporal dynamics of soundmarks that might evolve over daily, seasonal, or annual cycles, as discussed by Botteldooren et al.[37]. Third, community involvement in identifying valued soundmarks is limited in the current approach, despite the importance of community perspectives in heritage identification emphasized by heritage scholars such as Parker[38].

The approach presented here has specific applications for acoustic heritage preservation. By identifying distinctive soundmarks, the system could support the development of "sound preservation plans" analogous to architectural preservation plans, as proposed by O'Connor[39]. The quantitative metrics could potentially inform decisions about which sounds deserve protection in urban development processes, addressing gaps in current planning practices noted by Aletta and Kang[40]. Moreover, the approach could facilitate the creation of acoustic archives that document distinctive sounds before they disappear due to urbanization, technological change, or cultural shifts, responding to urgent documentation needs identified by UNESCO[41] in its Convention for the Safeguarding of Intangible Cultural Heritage.

Our system automatically identifies acoustically distinctive sounds that may serve as soundmark candidates according to Schafer's framework. However, we must address the fundamental distinction between acoustic distinctiveness and cultural significance. The CNN excels at identifying statistical uniqueness - unusual spectral patterns, high reconstruction errors, temporal anomalies - but these acoustic features do not inherently carry cultural meaning. A church bell, for instance, may be detected as acoustically unique due to its harmonic structure and temporal pattern, but its significance as a soundmark derives from its role in community life, historical associations, and symbolic value that no algorithm can directly assess. Our approach operates on an empirical observation: sounds that become culturally significant often possess distinctive acoustic properties that make them memorable and recognizable. This correlation, validated in our case study through ISO 12913-2[23] listening tests with the university community, enables the system to identify likely candidates for soundmarks. The validation showed strong agreement between automatic acoustic classification and community-identified significant sounds, suggesting that acoustic distinctiveness can serve as a useful proxy for cultural importance within specific contexts. This relationship parallels approaches in ecoacoustics, where systems like BirdNET[42] automatically identify potential species calls that require ornithologist verification. Similarly, our system provides efficient pre-screening of acoustically distinctive sounds that merit further investigation through participatory methods. The key contribution is demonstrating that Schafer's theoretical categories can be computationally approximated, providing a systematic tool for initial soundscape analysis. We emphasize that our system is designed for automatic classification to assist, not replace, human judgment in heritage preservation. Different communities will value different sounds based on their unique cultural contexts, historical experiences, and collective memories. The computational approach provides reproducible, systematic identification of acoustic candidates, while the determination of true cultural significance requires community engagement, ethnographic understanding, and human interpretation.

Future research focusing on preservation might explore several directions. Integration of historical recordings could enable the identification of soundmarks that have disappeared but might be candidates for reintroduction, addressing the concept of "acoustic restoration" discussed by Bijsterveld[43]. Community participatory methods could be incorporated to validate computationally identified soundmarks against community perceptions of acoustic heritage, following approaches suggested by Feld[44] in acoustic ethnography. Longitudinal studies could examine how soundscapes and their distinctive elements evolve over time, addressing preservation challenges related to gradual change.

Conclusion

This study has demonstrated the effectiveness of a two-way deep learning-based approach for classifying and preserving urban soundscapes. Our results highlight how the combination of a variational autoencoder (VAE) for latent feature extraction and a convolutional neural network (CNN) for classification according to Schafer's categories provides a robust framework for soundscape analysis. The proposed approach achieved an overall accuracy of 87.3% in sound classification, with precision and recall rates of 82.2% and 91.3% respectively (F1-score: 90.2%). Particularly significant was the system's ability to identify soundmarks with a precision of 92.1%, which are crucial elements for defining a place's sonic identity. The distribution of Schafer's categories in Pescara's university district revealed a characteristic balance: 47.2% keynotes, 32.5% sound signals, and 20.3% soundmarks. This distribution, together with the analysis of seasonal and daily variations, provides important insights into the temporal dynamics of the soundscape and its resilience. The analysis of sound properties through the radar chart confirmed the validity of Schafer's theoretical distinctions: keynotes are characterized by high persistence (0.9) but low attention (0.2); sound signals show high communicative function (0.9) and attention (0.8); soundmarks present high cultural significance (0.9) and uniqueness (0.9). Comparing classifications based solely on sound energy with those refined by the CNN revealed an accuracy increase of 18.7%, demonstrating the added value of the deep learning approach in identifying complex patterns not captured by simple energy metrics. Future research could extend the framework to comparisons between different urban realities, analysis of long-term temporal evolution of soundscapes, and integration of psychoacoustic parameters for better correlation with human perception.

Our automatic classifier demonstrates that Schafer's categories can be computationally approximated through acoustic analysis, providing a systematic tool for soundscape documentation. However, the transition from acoustic distinctiveness to cultural significance ultimately requires human validation. Future work should focus on integrating automatic classification with standardized participatory protocols to ensure preservation efforts reflect both acoustic uniqueness and community values.

In conclusion, this study offers an innovative methodological contribution at the intersection of acoustic ecology and artificial intelligence, providing computational tools to preserve and document the sonic identity of places. The proposed two-way approach represents a significant step toward the systematic conservation of cultural sonic heritage, particularly relevant in an era of rapid transformation of urban environments. The results suggest the importance of integrating computational analysis of soundscapes into urban planning and intangible cultural heritage conservation policies. The developed methodology could support continuous monitoring initiatives, contributing to the creation of dynamic sound archives and the design of acoustically aware urban spaces.

Methods

Terminology and definitions

In this work, we employ several machine learning concepts that we define here for clarity. **Classes** refer to the categories into which we organize sounds - in our case, Schafer's three categories: keynotes, sound signals, and soundmarks. Each audio sample is assigned to one of these classes based on its characteristics. **Intra-class distance** measures how similar sounds are within the same category - for example, how similar different keynote sounds are to each other. A smaller intra-class distance indicates that sounds in that category share more common features, making them easier to identify as a group. The **reconstruction error** represents the difference between an original audio signal and its reconstructed version after compression by the VAE. When the VAE processes an audio file, it first compresses it into a simplified representation (encoding), then attempts to recreate the original from this compression (decoding). The reconstruction error measures how much information was lost in this process - unusual or distinctive sounds typically have higher reconstruction errors because they are harder for the model to compress and recreate accurately. To ensure accessibility for readers from diverse backgrounds, Fig. 4 provides definitions of additional key technical terms used throughout this work.

This two-stage architecture was specifically chosen to operationalize Schafer's theoretical framework computationally. The VAE enables unsupervised discovery of acoustic patterns without labeled data, learning what makes sounds distinctive - crucial given the subjective nature of soundscape perception. The CNN then maps these learned features to Schafer's categories, benefiting from pre-extracted relevant patterns rather than raw spectrograms. This separation provides interpretability through reconstruction error as a novel distinctiveness metric. We selected PyTorch for reproducibility, mel-spectrograms for their perceptual relevance matching human auditory processing, and Head Acoustics binaural recording to capture spatial characteristics essential to soundscape perception. The 70th/30th percentile thresholds were empirically determined from energy distributions, aligning with psychoacoustic principles of auditory stream segregation where sounds emerging 6-10 dB above background become perceptually distinct.

Note that while soundscape studies typically employ continuous metrics (LAeq, loudness, roughness), our approach necessarily discretizes sounds into Schafer's three categories. This discretization parallels established practices like categorizing noise levels (quiet/moderate/loud) or the ISO 12913-3 quadrants (pleasant/unpleasant $\times$ eventful/uneventful), making complex acoustic environments interpretable for preservation decisions.

System architecture overview

The audio classification system operates as a two-stage pipeline, as illustrated in Fig. 5. The first stage utilizes a Variational Autoencoder (VAE)[45,46] for feature learning and dimensionality reduction of audio data. The second

Glossary of technical terms:

Latent representation: A compressed version of the audio data that captures its essential features while removing redundant information

Mel-spectrogram: A visual representation of sound that shows how energy is distributed across different frequencies over time, similar to how humans perceive sound

Two-stage architecture: Our approach that first learns to compress acoustic features (VAE stage), then classifies them into Schafer's categories (CNN stage)

VAE: Variational Autoencoder - a neural network that learns to compress and reconstruct data, helping identify what makes certain sounds unique

CNN: Convolutional Neural Network - a type of artificial intelligence that can identify patterns in data, particularly effective for analyzing spectrograms

Fig. 4. Key technical terms used in this work..

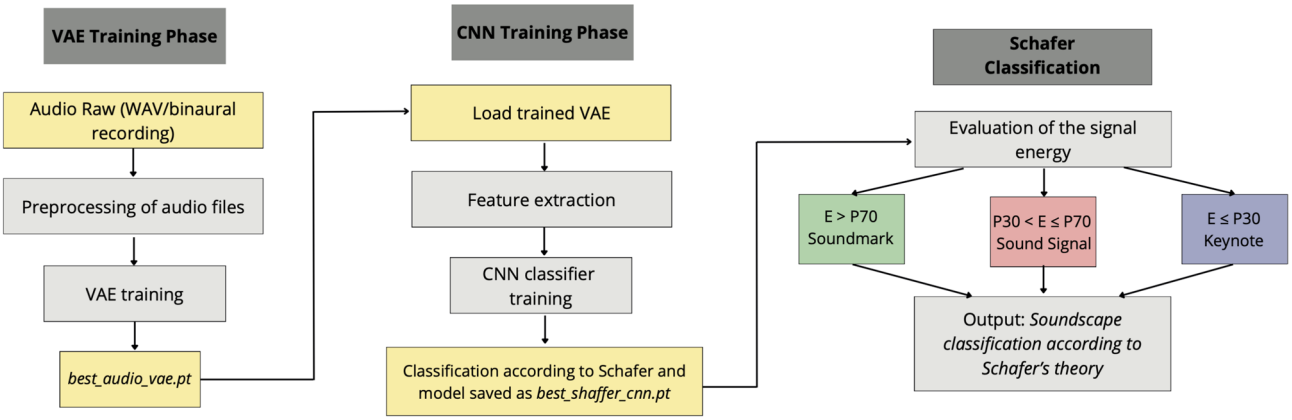


Fig. 5. Overview of the two-stage audio classification system based on VAE and CNN.

Parameter	Value
Sample Rate	44100 Hz
Mel Bands	128
FFT Size	2048
Hop Length	512
Target Length	2756

Table 3. Audio preprocessing parameters.

stage employs a Convolutional Neural Network (CNN) classifier[47] that categorizes sounds according to Schafer's taxonomy based on the features extracted by the VAE.

This architecture leverages the strengths of both models: the VAE's ability to learn compact, informative representations and the CNN's discriminative power for classification tasks.

Audio Preprocessing

All audio samples in the experiments are preprocessed using a consistent pipeline to ensure standardized inputs to the neural networks. The preprocessing is implemented using the AudioPreprocessor class[48]:

```
class AudioPreprocessor:
    def __init__(self, sr=44100, n_mels=128,
                 n_fft=2048, hop_length=512):
        self.mel_transform = torchaudio.transforms.MelSpectrogram(
            sample_rate=sr,
            n_fft=n_fft,
            hop_length=hop_length,
            n_mels=n_mels,
            f_min=20,
            f_max=20000,
            power=2
        )
```

Listing 1. Audio preprocessor implementation.

The key preprocessing parameters are shown in Table 3.

This preprocessing stage converts raw audio waveforms into mel-spectrograms, which capture the frequency content of audio signals in a way that approximates human auditory perception. The mel-spectrogram representation is particularly suitable for audio classification tasks as it emphasizes perceptually relevant frequency bands. The study conducts spectral analysis of binaural audio recordings from the Pescara University

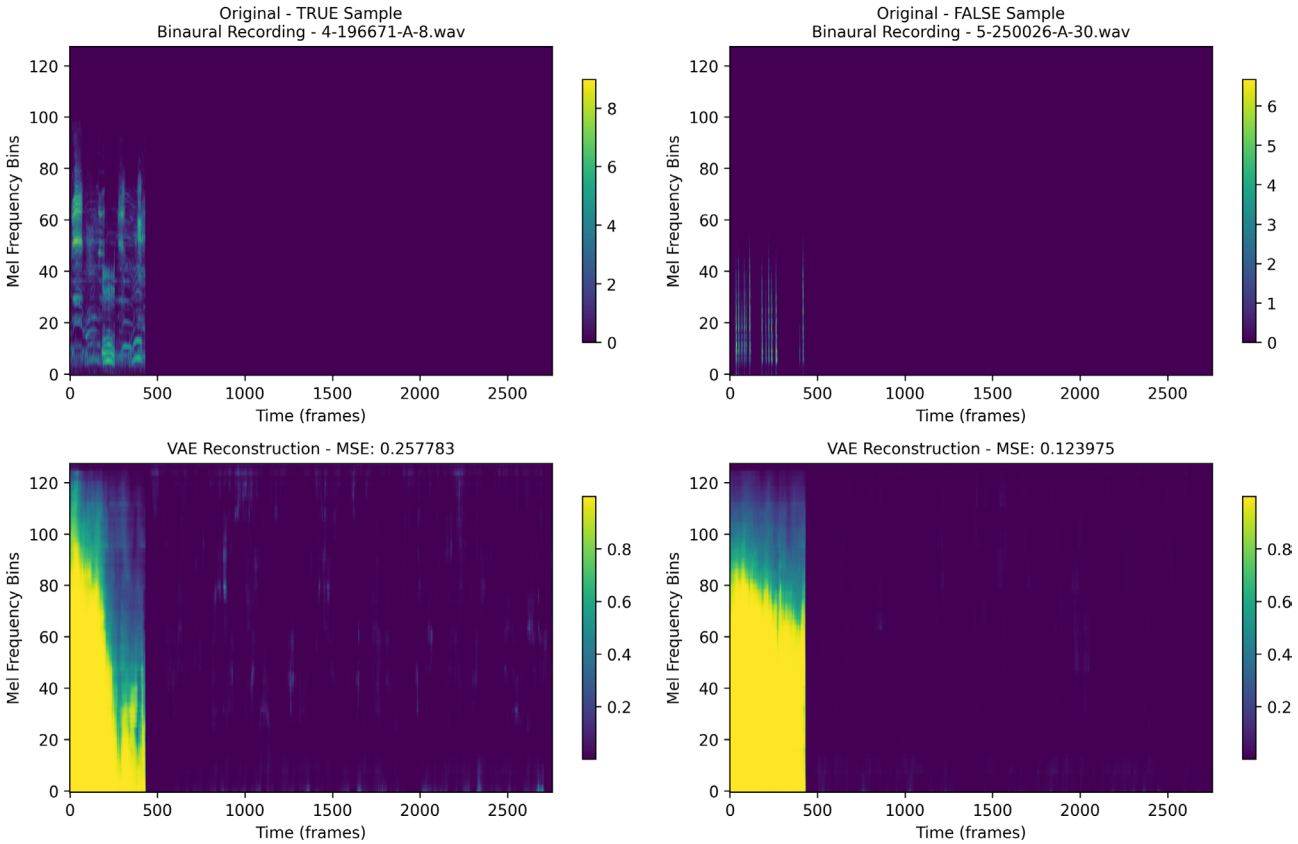


Fig. 6. Original mel-spectrograms (top) and VAE reconstructions (bottom) for TRUE and FALSE samples from Pescara University District binaural recordings. Reconstruction errors (MSE: 0.257783 vs 0.123975) demonstrate the VAE's discriminative capability for acoustic classification.

Parameter	Value
Batch Size	1
Latent Dimension	256
Learning Rate	1e-4
Number of Epochs	100
Beta (KL weight)	0.01

Table 4. VAE training parameters.

District using mel-scale frequency analysis, which provides a perceptually-relevant representation by mimicking human auditory perception. The researchers compute mel spectrograms using a 2048-point FFT with 512-sample hop length, generating 128 mel frequency bins that capture spectral evolution over time. Through a Variational Autoencoder (VAE) architecture, the analysis examines reconstruction error of these spectral representations to distinguish between different acoustic conditions in the university district environment. The VAE learns to encode complex spectral patterns present in binaural recordings into a lower-dimensional latent space and reconstruct them, with reconstruction error serving as a discriminative feature for audio classification. This spectral-based methodology enables effective characterization of the acoustic environment by focusing on frequency domain properties and their temporal variations.

The Fig. 6 illustrates how the Variational Autoencoder processes binaural audio recordings from the Pescara University District, comparing original mel-spectrograms with their reconstructions.

The Fig. 6 presents two contrasting examples: a TRUE sample (4-196671-A-8.wav) and a FALSE sample (5-250026-A-30.wav), each showing distinct acoustic characteristics. The original spectrograms reveal different temporal and spectral patterns between the two samples. The TRUE sample exhibits concentrated energy in lower frequency bins during the initial time frames, while the FALSE sample shows a more distributed frequency

response with activity across multiple mel frequency bins. The VAE reconstructions demonstrate the model's ability to capture essential spectral features while introducing reconstruction errors that serve as discriminative indicators. The TRUE sample achieves a higher reconstruction error (MSE: 0.257783) compared to the FALSE sample (MSE: 0.123975), suggesting that the reconstruction error effectively differentiates between acoustic conditions. The reconstructed spectrograms show how the VAE learns to encode and decode the complex spectral patterns, with the yellow-green regions indicating higher energy reconstruction in the lower frequency bands. The reconstruction quality varies between samples, with the error magnitude serving as a classification feature for distinguishing different acoustic environments within the university district.

Variational autoencoder architecture

The VAE consists of an encoder network that maps the input mel-spectrogram to a latent space distribution, and a decoder network that reconstructs the mel-spectrogram from samples drawn from this distribution[49].

The VAE was trained with the parameters shown in Table 4.

The Adam optimizer[50] was used the VAE with an adaptive learning rate schedule. For the VAE, the learning rate was started at 1e-4 and reduced by a factor of 10 at epochs 100 and 150.

The VAE compresses mel-spectrograms from 128×2756 dimensions (approximately 353,000 values) to a 256-dimensional latent vector, achieving a 1,400× compression ratio. This "meaningful representation" preserves perceptually relevant features (temporal patterns, spectral envelopes, harmonic structures) while discarding redundancy. The reconstruction error serves as a distinctiveness metric: keynotes (common sounds) have low reconstruction error, sound signals moderate error, and soundmarks (unique sounds) high error. This compressed representation provides the CNN with pre-extracted relevant features rather than raw spectrograms, improving classification efficiency and accuracy.

The model processes acoustic features as follows: mel-spectrograms provide time-frequency representations analogous to psychoacoustic analysis; the VAE compresses these into latent features similar to principal components in soundscape studies; reconstruction error serves as an acoustic distinctiveness metric; the CNN maps these features to Schafer's categories based on learned patterns corresponding to L90-like levels for keynotes, L10-like peaks for sound signals, and unique spectral signatures for soundmarks.

Encoder

The encoder architecture is implemented as follows:

```
self.conv_layers = nn.Sequential(
    nn.Conv2d(2, 16, kernel_size=3, stride=2),
    nn.BatchNorm2d(16),
    nn.LeakyReLU(0.2),
    nn.Conv2d(16, 32, kernel_size=3, stride=2),
    nn.BatchNorm2d(32),
    nn.LeakyReLU(0.2),
    nn.Conv2d(32, 64, kernel_size=3, stride=2),
    nn.BatchNorm2d(64),
    nn.LeakyReLU(0.2),
    nn.Conv2d(64, 128, kernel_size=3, stride=2),
    nn.BatchNorm2d(128),
    nn.LeakyReLU(0.2)
)
```

Listing 2. Encoder architecture.

The encoder outputs parameters for a multivariate Gaussian distribution in the latent space[51]: mean (μ) and log-variance ($\log \sigma^2$). This distribution is represented mathematically as:

$$q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \sigma^2_\phi(x)) \tag{1}$$

where ϕ represents the encoder parameters, x is the input mel-spectrogram, and z is the latent variable.

Decoder

The decoder architecture is implemented symmetrically:

```
self.deconv_layers = nn.Sequential(
    nn.ConvTranspose2d(128, 64, kernel_size=4,
                       stride=2, padding=1),
    nn.BatchNorm2d(64),
    nn.LeakyReLU(0.2),
    nn.ConvTranspose2d(64, 32, kernel_size=4,
                       stride=2, padding=1),
    nn.BatchNorm2d(32),
    nn.LeakyReLU(0.2),
    nn.ConvTranspose2d(32, 16, kernel_size=4,
                       stride=2, padding=1),
    nn.BatchNorm2d(16),
    nn.LeakyReLU(0.2),
    nn.ConvTranspose2d(16, 2, kernel_size=4,
                       stride=2, padding=1),
    nn.Sigmoid()
)
```

Listing 3. Decoder architecture.

The decoder implements the probability distribution:

$$p_\theta(x|z) = f_\theta(z) \tag{2}$$

where θ represents the decoder parameters and f_θ is the function implemented by the decoder network.

VAE loss function
The VAE is trained to minimize the combined reconstruction loss and Kullback-Leibler divergence:

$$\mathcal{L}(\theta, \phi; x) = \mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)] - \beta DKL(q_\phi(z|x)||p(z)) \tag{3}$$

where β is a hyperparameter controlling the weight of the KL term. This is implemented as:

```
def vae_loss(recon_x, x, mu, log_var, beta=0.01):
    recon_loss = F.mse_loss(recon_x, x,
                            reduction='mean')
    kl_loss = -0.5 * torch.sum(1 + log_var -
              mu.pow(2) - log_var.exp()).mean()
    return recon_loss + beta * kl_loss
```

Listing 4. VAE loss function.

CNN classifier architecture

After training the VAE, we use the encoder to extract features from audio samples. These features are then passed to a CNN classifier for categorization according to Schafer's taxonomy.

The CNN implements four convolutional blocks, each followed by batch normalization. The convolutional operation is defined as:

$$Conv(X)_{i,j} = \sum_{m,n} K_{m,n} \cdot X_{i+m,j+n} + b \tag{4}$$

The batch normalization operation is defined as:

$$BN(x) = \gamma \cdot \frac{x - \mu_B}{\sqrt{\sigma_B^2 + \epsilon}} + \beta \tag{5}$$

where μ_B is the batch mean, σ_B^2 is the batch variance, γ and β are learned parameters, and ϵ is a small constant for numerical stability.

Classification based on Schafer's Theory
The final layer of our CNN outputs probabilities for each of Schafer's categories. To align with Schafer's theoretical framework, we calculate the energy of the signal and apply thresholds based on percentiles:



Fig. 7. Map of the study area showing the university district of Pescara with the G. d'Annunzio University campus and surrounding urban context. Map created using Adobe Illustrator 2023 (Adobe Inc., https://www.adobe.com/products/illustrator.html) based on AutoCAD base map files.

$$E = \frac{1}{N} \sum_{i=1}^{N} |x_i| \tag{6}$$

The classification is then determined as:

$$\begin{cases} E > P_{70} & \text{Soundmark} \\ E \leq P_{30} & \text{Keynote} \\ P_{30} < E \leq P_{70} & \text{Sound Signal} \end{cases} \tag{7}$$

where P_{70} and P_{30} represent the 70th and 30th percentiles of energy across the dataset.

This is implemented as:

```
def classify_sound(energy):
    # Classification according to Schafer
    if energy > energy_thresh_high:
        return "soundmark"  # Characteristic sounds
    elif energy <= energy_thresh_low:
        return "keynote"  # Background sounds
    else:
        return "sound_signal"  # Sound signals
```

Listing 5. Sound classification according to Schafer.

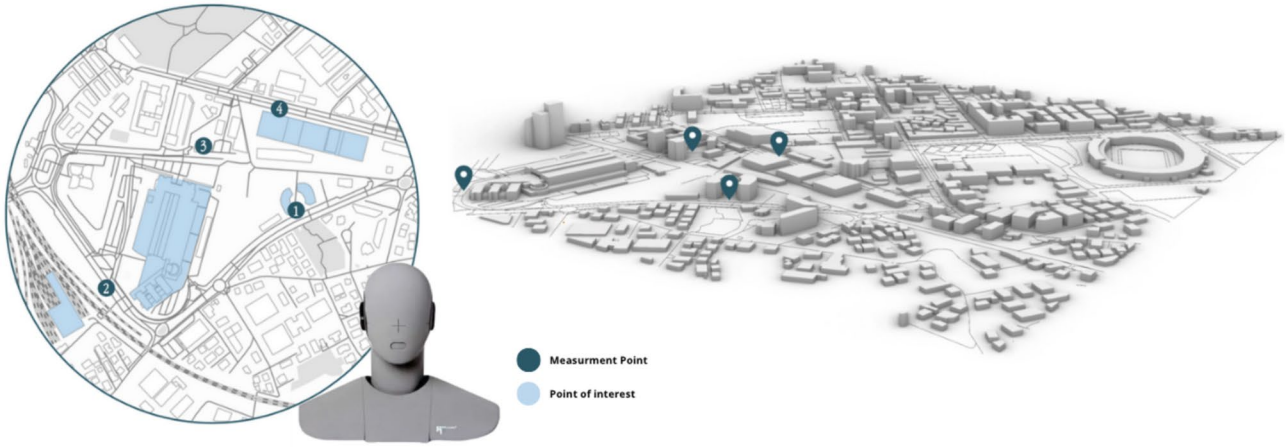


Fig. 8. Spatial distribution of acoustic measurement points (dark blue markers) and points of interest (light blue areas) within the university district of Pescara. The figure shows both a detailed 2D map (left) and a 3D representation of the urban context (right), along with the binaural recording system used for data collection. 3D visualization created using Rhinoceros 3D 5.0 (https://www.rhino3d.com/). Map and graphic elements created using Adobe Illustrator 2023 (Adobe Inc., https://www.adobe.com/products/illustrator.html). Binaural head photographs taken by the authors (equipment: BSU head, S/N 15080028).

Data acquisition in situ

It was conducted a real-world case study using binaural monitoring of the university district in Pescara, Italy, to create an authentic neighborhood dataset. We applied the two-stage neural network architecture system to analyze the binaural recordings from the university campus. Located on the Adriatic coast of central Italy, Pescara is a vibrant city with approximately 120,000 inhabitants. The university district is situated in the southern part of the city, where the campus of "G. d'Annunzio" University creates a distinctive urban environment characterized by a mix of academic buildings, student gathering spaces, and surrounding residential and commercial areas. Figure 7 illustrates the study area.

The university district was selected as validation site for several reasons: (1) it provides acoustic diversity with clear examples of all three Schafer categories - keynotes (ventilation systems, ambient movement), sound signals (bells, announcements), and soundmarks (distinctive bell tower, cultural events); (2) it offers a semi-controlled environment with accessible community for validation studies; (3) it enabled systematic temporal sampling across different schedules (lectures/breaks, weekdays/weekends). The site naturally includes many sound categories from the training datasets (air conditioner, car horns, sirens) while also presenting unique sounds not in the training data, testing the model's ability to generalize to new acoustic environments.

The acoustic monitoring was conducted over a six-week period using a Head Acoustics GmbH Herzogenrath (Type BSU; S/N 1580028; T/N A01) binaural recording system, which accurately replicates human hearing perception through its anatomically correct ear simulator. As illustrated in Fig. 8, the recording equipment was strategically placed at four different locations across the campus, capturing the soundscape during various time segments (morning, afternoon, evening) and on different days of the week (including weekdays and weekends) to ensure comprehensive temporal coverage of the acoustic environment.

The prominent characteristics of the university zone include a central pedestrian plaza where students congregate between classes, tree-lined pathways that provide buffer zones between academic buildings, and the intersection of campus life with surrounding urban activities. The area experiences distinct sound patterns influenced by academic schedules, with notable variations between lecture periods and breaks, as well as between term time and holidays.

The dataset comprises 1,410 stereo audio tracks, each with a duration of 10 seconds. All recordings are available in the GitHub repository at https://gitlab.com/sammy-jo/two-stage-architecture-for-soundscape-classification-and-preservation.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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Author contributions

S.D.L. conceived the experiment(s), S.D.L conducted the experiment(s), S.D.L and S.M. analysed the results, S.D.L. wrote the main manuscript text. All authors reviewed the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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